Lorenzo Veronese

Curriculum Vitae

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Profile

PhD Candidate at Politecnico di Milano, specializing in Computational Cognitive Neuroscience and Neuro-inspired AI. My research focuses on developing state-of-the-art AI models, grounded in fMRI data, to emulate complex human functions such as planning (through theory-driven AI models informed by neuroscience, psychology, and neuropsychology) and visual perception (via advanced two-stage generative decoding architectures). Published work includes enhancements to neural decoding pipelines, demonstrating superior performance and efficiency (more than 95% improvement) for real-time applications.

Education

2024–Present **Ph.D., Bioengineering**, *Politecnico di Milano (Milan, Italy)*, Supervisors: Prof. Pietro Cerveri, Dr. Pasquale Anthony Della Rosa, Computational Cognitive Neuroscience for Understanding Human Planning and Perception.

2022–2024 **M.Sc., Computer Science Engineering, Artificial Intelligence**, *Politecnico di Milano (Milan, Italy)*, Thesis: "Visual stimulus reconstruction from BOLD fMRI signal using generative AI", Final grade: 110 *cum laude*/110 (GPA: 29.2).

2019–2022 **B.Sc., Computer Science Engineering**, *Politecnico di Milano (Milan, Italy).*

Research Experience

2024–Present **Research fellow and PhD student**, *Politecnico di Milano (Department of Electronics, Information and Bioengineering) and IRCCS Ospedale San Raffaele (Department of Neuroradiology).*

- Neuro-AI Generative Modeling: Designing an explainable two-stage generative decoding pipeline to reconstruct visual perception from fMRI data.
- Computational Cognitive Neuroscience: Formulating a theory-driven probabilistic model of human planning in the Tower of London neuropsychological task.

2021–2023 **Student Researcher**, *Politecnico di Milano (Department of Electronics, Information and Bioengineering), NECSTLab.*

- Generative AI for Data Harmonization: Engineered a CycleGAN-based deep learning framework for the harmonization of histopathological images across different imaging domains.

2022–2023 **Student Researcher**, *Politecnico di Milano (Department of Electronics, Information and Bioengineering), DEEP-SE Lab.*

- Formal Modeling and Natural Language Processing: Modeled the dynamics of psychotherapy conversations using Stochastic Hybrid Automata and natural language processing.

2021–2022 **Student Researcher**, *Politecnico di Milano (Department of Electronics, Information and Bioengineering), AIRLab.*

- Robotic Design and Prototyping: Led the full-stack design (mechanical and software) of an assistive robotic device to support individuals with motor disabilities in writing tasks.

Scientific Publications

Dettori, F.*, Forasassi, M.*, **Veronese, L.***, Lestingi, L., Scotti, V., & Rossi, M.G. (2026). Do You Understand How I Feel?: Towards Verified Empathy in Therapy Chatbots. *ArXiv.* 10.48550/arXiv.2601.08477. (*Equal contribution)

Veronese, L., Pecco, N., Moglia, A., Scifo, P., Castellano, A., Mainardi, L., Della Rosa, P.A., & Cerveri, P. (2025). Evaluating Semantic Brain Regions Contribution to Visual Neural Decoding Performance on Different Classes of Stimuli. *20th conference on Computational Intelligence methods for Bioinformatics and Biostatistics (CIBB 2025).* (Oral presentation)

Veronese, L., Moglia, A., Pecco, N., Della Rosa, P.A., Scifo, P., Mainardi, L., & Cerveri, P. (2025). Optimized AI-based neural decoding from BOLD fMRI signal for analyzing visual and semantic ROIs in the human visual system. *Journal of Neural Engineering*, 22(4). 10.1088/1741-2552/adfbc2.

Veronese, L., Poles, I., D'Arnese, E., Santambrogio, M.D. (2023). Stain Transfer using CycleGAN for Histopathological Images. *IEEE EUROCON 2023-20th International Conference on Smart Technologies.* 10.1109/EUROCON56442.2023.10199027.

Teaching

2025 **Teaching Assistant, Neuroengineering**, *Politecnico di Milano*

- Delivered lectures on deep learning architectures and methodologies.
- Supervised student groups in designing and training end-to-end deep learning models for decoding visual categories from fMRI data.
- Co-designed course materials and organization.

2025 **Teaching Assistant, Fundamentals of Computer Science**, *Politecnico di Milano*

- Led laboratory sessions for undergraduate students, teaching algorithmic thinking and data structures in C programming.
- Co-designed course materials.

Awards & Honors

2025 Selected by Nova talent (ID: 6698IITXRXG57SY19CHV5KD3E) — Nova

2022 Scholarship for academic merit — Town of Cucciago & Rubelli

Advanced Training & Summer Schools

2025 **European Course on Advanced Imaging Techniques in Neuroradiology**, *IRCCS Ospedale San Raffaele*, Milan, Italy

2025 **XLIV Annual School of Bioengineering**, *University of Pisa*, Pisa, Italy
Topic: "Unlocking the Mind and Emotions: Cutting-Edge Bioengineering Techniques from Computational Physiology to Clinical Applications"

Skills

Expertise	Computational Cognitive Neuroscience, Artificial Intelligence, Machine Learning, Deep Learning, fMRI, Cognitive Neuroscience, Public Speaking
Programming	Python, PyTorch, TensorFlow, MATLAB, Bash
Tools	Git, LaTeX, Statistical Parametric Matching (SPM), FSL
Languages	English (fluent), Italian (native)

References

Prof. Pietro Cerveri	Full Professor — Università di Pavia pietro.cerveri@unipv.it
Dr. Pasquale Anthony Della Rosa	Scientific Collaborator — IRCCS Ospedale San Raffaele dellarosa.pasquale@hsr.it

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